Financial Access and the Local Resource Curse: Evidence from Mining Transfers in Peru

José Mendoza*

Pontificia Universidad Católica del Perú

April 2026

Abstract

Why do resource windfalls stimulate growth in some places but not others? Using a district-level panel combining fiscal records, banking data, and satellite nighttime lights for 1,800 Peruvian districts over 2012–2024, I estimate the fiscal multiplier of mining canon transfers instrumented with a Bartik design. In districts without financial institutions, exogenous fiscal windfalls are associated with reduced nighttime light intensity ($\hat{\beta}_1 = -0.123$, $p < 0.01$), consistent with a local resource curse. The interaction between spending and financial access is positive and of similar magnitude ($\hat{\beta}_3 = 0.126$, $p < 0.01$), though this estimate attenuates to 0.07–0.08 when controlling for the interaction of spending with baseline development level, suggesting that financial access partially proxies for broader district characteristics. The complementarity is similar across commercial banks, municipal savings banks, and the state-owned Banco de la Nación—a pattern whose interpretation remains an open question, as Banco de la Nación presence may proxy for state capacity rather than financial intermediation per se.

*Department of Economics, Pontificia Universidad Católica del Perú (PUCP). Email: jose.mendozas@pucp.edu.pe

Keywords: resource curse; fiscal transfers; financial intermediation; mining canon; nighttime lights; Peru

JEL codes: O13, O16, H72, Q32, R11

1 Introduction

Over fifty countries redistribute revenues from natural resource extraction to subnational governments, expecting these fiscal windfalls to promote local development (Venables, 2016). The logic is intuitive: extractive industries generate rents that, if shared with producing regions, can finance infrastructure, public services, and the foundations of long-run growth. Yet a striking regularity in the empirical record is that resource-rich subnational jurisdictions rarely outperform their neighbors. In Brazil, municipalities receiving oil royalties show no improvement in public goods provision despite dramatically larger budgets (Caselli and Michaels, 2013). In Peru, districts that received the largest mining transfers during the 2004–2012 commodity boom saw no clear reduction in poverty (Loayza and Rigolini, 2016), and in some cases the transfers exacerbated local conflict (Arellano-Yanguas, 2011). The pattern is sufficiently robust to have earned its own label: the "local resource curse" (van der Ploeg, 2011).

What makes this puzzle especially sharp is that the same fiscal transfer can produce very different outcomes depending on where it lands. Some recipient districts grow; others stagnate or decline. Previous explanations have focused on governance failures—corruption (Brollo et al., 2013), low spending quality (Ardanaz and Maldonado, 2023), or institutional weakness more broadly. These are surely important, but they raise a prior question: even in the absence of corruption, through what mechanism does a peso of municipal spending become a peso of local economic activity? If a district government hires a contractor to pave a road, the contractor pays workers, who buy goods from local shops, who receive payments through the banking system. This circulation requires, at minimum, that the local economy have the financial infrastructure to process and retain money. In a district with no bank, no savings institution, and no formal payment system, much of the spending leaks out immediately—through cash that travels to the nearest town, purchases from non-local suppliers, or savings held under the mattress.

This paper tests a simple hypothesis: the presence of financial institutions moderates the effect of fiscal windfalls on local economic activity. I study Peru's mining canon—a program through which 50% of corporate income tax from mining is distributed to subnational governments based on geographic proximity to mines. Between 2004 and 2024, Peru distributed over 80 billion soles (approximately USD 25 billion) through this mechanism, making it one of the largest subnational revenue-sharing programs in Latin America. The setting is ideal for two reasons. First, canon transfers are highly volatile because they track international commodity prices with a short lag, generating large and plausibly exogenous fiscal shocks. Second, financial access varies sharply across Peru's approximately 1,900 districts: in 2012, only 26% had even a single bank branch or savings institution office.

I construct a novel district-level panel combining four data sources: fiscal records from Peru's integrated financial management system (SIAF), financial institution presence from the banking regulator (SBS), satellite nighttime lights from the VIIRS sensor, and international commodity prices from the World Bank. The estimation strategy employs a Bartik instrument—the interaction of a district's pre-determined canon dependence (measured in 2012) with an index of international mineral prices—to generate exogenous variation in municipal spending driven by global commodity markets.

The main finding, based on the 2017–2024 commodity recovery period, is stark. In districts without financial institutions, a one-unit increase in log municipal spending *reduces* nighttime light intensity by 0.123 log points ($p < 0.01$), consistent with a local resource curse. But the interaction between spending and financial access is positive and nearly identical in magnitude ($\hat{\beta}_3 = 0.126$, $p < 0.01$), fully offsetting the negative effect. In districts with at least one financial institution, the net fiscal multiplier is approximately zero. The Kleibergen-Paap F-statistic is 32.7, well above conventional thresholds for instrument strength (Stock and Yogo, 2005).

A natural concern is that commodity prices affect local economies directly through mining activity, not just through the fiscal channel. I address this by excluding the 334 districts that host active mining concessions, identified from INGEMMET's official mining cadastre. The complementarity result survives ($\hat{\beta}_3 = 0.101$, $p < 0.01$), confirming that identification comes from the fiscal channel—canon transfers that flow to districts based on geographic formulas, many of which host no mines at all.

The complementarity is remarkably consistent across institution types. Commercial banks ($\hat{\beta}_3 = 0.089$), municipal savings banks ($\hat{\beta}_3 = 0.091$), and the state-owned Banco de la Nación ($\hat{\beta}_3 = 0.091$) all show similar moderating effects, each significant at the 1% level. That the state-owned Banco de la Nación—which primarily processes government payments rather than extending credit—shows a similar effect to commercial banks is suggestive but difficult to interpret causally: Banco de la Nación presence is a powerful proxy for state capacity, formalization, and integration into the national economy, and its moderating effect could reflect these correlated characteristics rather than its specific function as a payment platform.

These findings contribute to three literatures. First, I add to research on the subnational resource curse (Caselli and Michaels, 2013; Aragón and Rud, 2013; Loayza and Rigolini, 2016) by identifying financial access as a moderating mechanism—related to but distinct from the institutional quality channel emphasized by Mehlum et al. (2006)—shifting the focus from governance failures to absorptive capacity. Second, I provide subnational causal evidence for the hypothesis, advanced at the cross-country level by Bhattacharyya and Hodler (2014), that financial development moderates the resource curse—connecting the resource curse literature to the broader finance-and-growth tradition (King and Levine, 1993; Rajan and Zingales, 1998; Beck et al., 2000). Third, I contribute to the sparse evidence on fiscal multipliers in developing countries (Corbi et al., 2019; Gadenne, 2017; Nakamura and Steinsson, 2014; Chodorow-Reich, 2019), showing that the multiplier is *conditional* on local financial infrastructure—a finding with direct implications for the design of fiscal decentralization.

The paper proceeds as follows. Section 2 describes Peru's canon system, the data sources, and the construction of key variables. Section 3 develops the empirical strategy. Section 4 presents results, robustness checks, and threats to identification. Section 5 concludes.

2 Setting and Data

Peru is one of the world's largest producers of copper, gold, zinc, and silver, with mining contributing approximately 10% of GDP and over 60% of export revenues. The fiscal link between mining and local governments runs through the canon minero, established by Ley 27506 (2001), which earmarks 50% of corporate income tax paid by mining companies for distribution to subnational governments in the regions where extraction occurs.

The distribution formula is geographic rather than production-based: 10% goes to the district where the mine is located, 25% to the province, 40% to the department, and 25% to the regional government (which must allocate 20% of its share to public universities in the region). This means that many districts receiving substantial canon transfers do not themselves host mines—they simply lie in a mining province or department. The geographic diffusion is important for identification: a district's canon dependence reflects its *fiscal* reliance on mining revenues, not its direct exposure to mining activity. Canon transfers are earmarked for capital expenditure, though evidence suggests substantial fungibility in practice (Ardanaz and Maldonado, 2023).

The magnitude of canon varies enormously across districts. In the sample, the median district derives approximately 43% of its modified institutional budget from canon, but this ranges from under 1% to over 96%. Critically, canon revenue is highly volatile: it tracks international commodity prices with a one- to two-year lag due to the corporate tax assessment cycle. Figure 1 illustrates the strong co-movement between the mining price index and mean canon revenue across districts. The commodity bust of 2012–2016 halved canon transfers in many districts; the subsequent recovery through 2024 restored—and in some cases exceeded—earlier levels.

Financial inclusion has expanded in Peru since the early 2000s, driven by three types of institutions with distinct geographic footprints. The Banco de la Nación, a state-owned bank, operates branches in most provincial capitals and serves primarily as a platform for government payments—pensions, transfers, and public-sector salaries. Cajas municipales and cajas rurales, municipal and rural savings banks, have

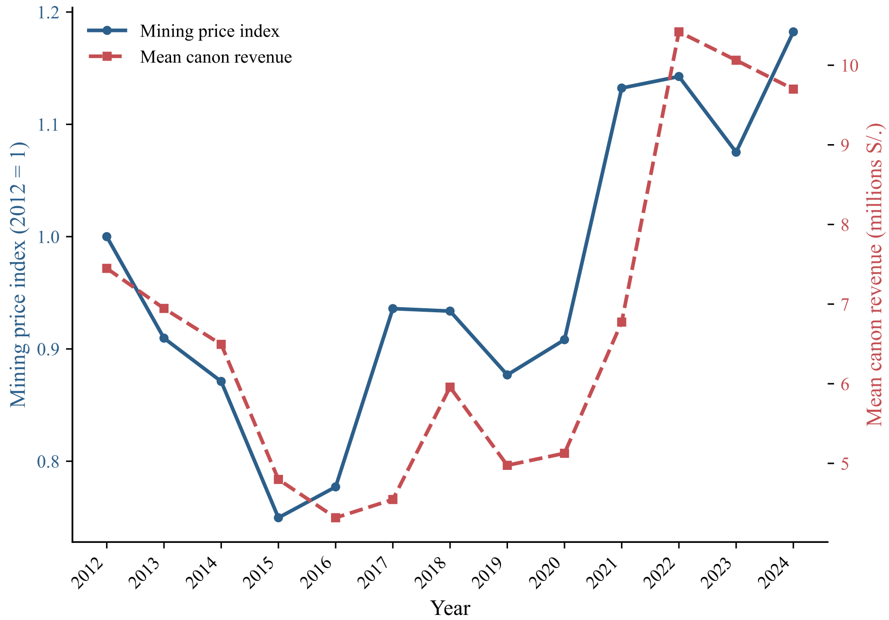

Figure 1: Mining price index and mean canon revenue, 2012–2024. The mining price index (left axis) is a weighted average of copper, gold, zinc, silver, and lead prices, normalized to 2012 = 1. Mean canon revenue (right axis) is the cross-district average of canon minero transfers in millions of soles.

expanded aggressively into secondary towns, specializing in small-business and agricultural lending. Commercial banks (banca múltiple) remain concentrated in urban areas. In 2012, approximately 26% of districts had at least one financial institution office; by 2024, this had risen to 29%, with 59 districts gaining access during the panel period. Financial presence is highly persistent: once a branch opens, it rarely closes, reflecting the high fixed costs of rural operations.

I construct a panel of 1,893 districts observed annually from 2012 to 2024, combining four data sources. The start date is determined by the availability of VIIRS nighttime lights data (beginning April 2012). After merging and dropping observations with missing values on key variables, the estimation sample contains approximately 23,400 district-year observations covering 1,799 districts.[1]

Nighttime lights. I measure local economic activity using annual mean radiance from the VIIRS sensor aboard the Suomi NPP satellite. Relative to the older DMSP-

[1]Of the 1,893 districts in the panel, 94 lack nighttime lights data (corresponding to districts created after GADM compiled its administrative boundaries), reducing the estimation sample to 1,799. The exact sample size varies slightly across specifications: the full-sample IV has $N = 23{,}361$, the post-2017 preferred specification has $N = 14{,}376$, and the first stage (which does not require nighttime lights) has $N = 23{,}862$.

OLS system, VIIRS offers substantially higher spatial resolution and a wider dynamic range, reducing saturation in urban areas and improving detection in rural ones (Gibson et al., 2021; Chen and Nordhaus, 2011; Michalopoulos and Papaioannou, 2013). I extract district-level radiance from the NOAA/VIIRS/DNB/MONTHLY_V1/VCMCFG monthly composites via Google Earth Engine, averaging monthly images within each calendar year and computing the mean within each district's GADM Level 3 boundaries. To handle zero and negative values arising from background noise subtraction, I apply the inverse hyperbolic sine (asinh) transformation, which approximates $\log(2x)$ for large x but is defined at zero (Henderson et al., 2012). Matching GADM boundaries to INEI district identifiers (ubigeos) required a multi-step crosswalk that resolved 1,803 of 1,815 polygons (99.3%); the 12 unmatched correspond to water bodies.

Fiscal data. Municipal spending and revenue data come from SIAF via MEF's Consulta Amigable portal. The main spending variable is *devengado*—the accrual-based measure that captures obligations recognized by the government regardless of cash disbursement. I systematically exclude entries for *mancomunidades municipales* (municipal consortia), which share the same district identifier as the host municipality and, if not filtered, distort budget share calculations. The endogenous variable is $\log G_{dt}$, log total municipal expenditure. For the Bartik instrument, the canon share is $s_d = \text{canon\_PIM}_{d,2012}/\text{gasto\_total\_PIM}_{d,2012}$, the ratio of canon revenue to total budget in the initial year (see Figure 4 in the Appendix for the distribution of s_d). The mining price index m_t is a weighted average of normalized commodity prices (copper, gold, zinc, silver, lead) from the World Bank Pink Sheet, with 2012 = 1.

Financial access. The SBS maintains a comprehensive registry of all financial institution offices by district and year. The main variable is $F_{dt} = \mathbb{1}[\text{N. offices}_{dt} > 0]$, an indicator for whether the district has at least one office. I also construct institution-type-specific indicators for commercial banks (BM, *banca múltiple*), cajas municipales (CM), and the Banco de la Nación (EE, *empresas estatales*, the SBS classification for state-owned financial institutions).

Table 1 reports summary statistics for the estimation sample. The median district

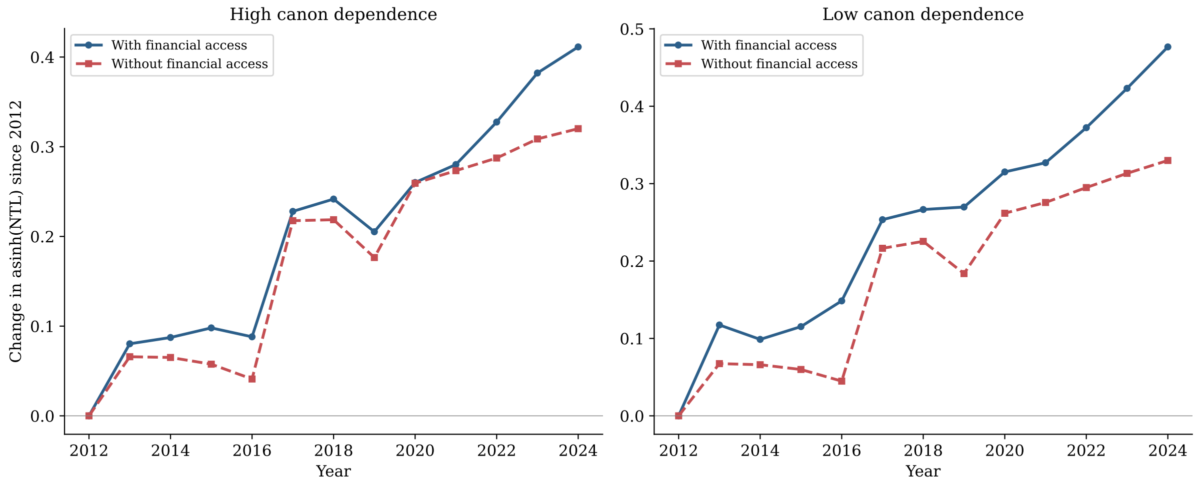

Figure 2: Change in nighttime lights since 2012, by canon dependence and financial access. Districts are split at the median of the pre-period canon share. Lines show the change in mean asinh(NTL) relative to 2012 for districts with financial access (solid blue) and without (dashed red). The divergence after 2018—particularly in the left panel—coincides with the commodity price recovery that drives the Bartik instrument.

derives 39% of its budget from canon, with substantial cross-sectional variation. Financial access is present in 28% of district-year observations, though this masks considerable heterogeneity: most districts either always have a bank (488) or never do (1,346), with only 59 switchers during the panel.

Figure 2 provides a first look at the key relationship. Splitting districts by canon dependence (above/below median) and financial access, and normalizing to 2012 levels, reveals a suggestive pattern: in both panels, districts with financial institutions (solid lines) experienced faster growth in nighttime lights than those without (dashed lines). The gap widens after 2018, coinciding with the commodity price recovery that generates the fiscal shocks driving the instrument. The econometric analysis that follows tests whether this pattern reflects a causal relationship.

3 Empirical Strategy

The goal is to estimate how financial access moderates the causal effect of municipal spending on local economic activity. The main specification is:

$$\text{asinh}(\text{NTL}_{dt}) = \alpha_d + \delta_t + \beta_1 \log G_{dt} + \beta_2 F_{dt} + \beta_3 (\log G_{dt} \times F_{dt}) + \varepsilon_{dt} \quad (1)$$

Table 1: Summary Statistics

	Mean	SD	Median	Min	Max
Panel A: Outcome					
Nighttime lights (mean radiance)	1.501	7.050	0.242	-0.059	100.619
Nighttime lights (asinh)	0.450	0.784	0.239	-0.059	5.305
Panel B: Fiscal variables					
Municipal spending (log)	15.601	1.255	15.542	12.565	21.652
Municipal spending (million soles)	14.84	47.87	5.62	0.29	2,530.87
Canon revenue (million soles)	6.52	24.48	1.70	0.00	1,188.95
Canon share (2012)	0.447	0.237	0.434	0.005	0.968
Panel C: Instrument					
Mining price index (2012 = 1)	0.961	0.132	0.934	0.750	1.182
Bartik instrument	0.429	0.237	0.413	0.004	1.144
Panel D: Financial access					
Has financial institution (=1)	0.277	0.448	0.000	0.000	1.000
Number of offices	2.361	9.159	0.000	0.000	125.000

Full sample: $N = 23{,}361$ district-year observations, 1,799 districts, 2012–2024. Preferred post-2017 sample: $N = 14{,}376$ (descriptives are similar; mean NTL rises from 0.45 to 0.54, mean log spending from 15.6 to 15.7).

where α_d and δ_t are district and year fixed effects, G_{dt} is total municipal expenditure, and F_{dt} is the financial access indicator. The coefficient β_1 captures the fiscal multiplier in districts without financial institutions, β_3 measures the additional effect when financial access is present, and $\beta_1 + \beta_3$ gives the multiplier in districts with financial institutions. Standard errors are clustered at the district level throughout.

The key identification challenge is that municipal spending is simultaneously determined with local economic conditions: more active districts generate more own-source revenue and may attract additional transfers. Moreover, canon transfers depend on mining profits, which could affect local economies directly through employment and procurement. To isolate exogenous variation in spending, I use a Bartik-style instrument:

$$Z_{dt} = s_d \times m_t \tag{2}$$

where s_d is the canon share measured in the pre-period (2012) and m_t is the international mining price index. The instrument interacts a time-invariant, cross-sectional exposure measure with a time-varying, aggregate shock. Conditional on district and year fixed effects, it exploits the fact that districts with higher pre-determined canon dependence experience larger fiscal shocks when international mineral prices fluctuate. Following the taxonomy of Borusyak et al. (2022), the design relies on the exogeneity of the shocks (m_t, international mineral prices) rather than the shares (s_d, canon dependence), since the latter are correlated with pre-existing district characteristics as shown in Figure 3(a). Under the Goldsmith-Pinkham et al. (2020) framework, which requires share exogeneity, the declining level effects visible in the event study would be a concern; I adopt the shock-exogeneity perspective, where the identifying assumption is that international commodity prices are orthogonal to idiosyncratic district-level shocks conditional on fixed effects.

The exclusion restriction requires that international mineral prices affect local economic activity only through their effect on canon-funded spending, conditional on district and year fixed effects. Year fixed effects absorb any aggregate effect of commodity prices on the Peruvian macroeconomy—exchange rates, national fiscal policy, overall demand. The remaining concern is that mineral prices differentially affect specific districts through direct mining activity. I address this in three ways: first, the exposure variable captures fiscal dependence, not production intensity, and many high-canon districts receive transfers through provincial and departmental formulas without hosting mines; second, I show robustness to excluding the 334 districts with active mining concessions; third, I include an interaction of a mining-district indicator with year fixed effects, allowing mining and non-mining districts to follow separate time trends. A residual concern is that non-mining districts located in mining-intensive provinces may still be exposed to regional spillovers—labor market commuting, supply chain linkages, and local price effects—that the district-level exclusion restriction does not fully address. To the extent that these regional spillovers correlate with the canon share instrument, the exclusion restriction is weakened.

I treat financial access as exogenous in equation (1). Two observations support this: the correlation between F_{dt} and the Bartik instrument is -0.08, essentially zero, and financial institution location decisions are driven by population density and urbanization rather than commodity price fluctuations. Accordingly, I instrument only

$\log G_{dt}$ using Z_{dt} as the excluded instrument, while F_{dt} and the interaction $\log G_{dt} \times F_{dt}$ enter as included regressors. This is a single-endogenous-variable setup: the 2SLS first stage regresses $\log G_{dt}$ on Z_{dt}, F_{dt}, and $\log G_{dt} \times F_{dt}$, and the fitted values replace $\log G_{dt}$ in the second stage. The Kleibergen-Paap F-statistic of 32.7 refers to this single first stage.[2]

Two threats to identification deserve explicit acknowledgment. First, the canon share is correlated with pre-existing nighttime light trends: more canon-dependent districts experienced declining luminosity even before the sample period, consistent with broader structural differences between mining-dependent and non-mining-dependent areas. Second, the *differential* pre-trend—whether the canon-dependence–financial-access interaction predicts nighttime light trends in the pre-period—is statistically significant ($F = 5.09$, $p = 0.002$) in the full-sample event study. This is the primary reason I focus on the 2017–2024 post-recovery sample as the preferred specification. In the post-2017 event study (with 2017 as base year), the first lead has a coefficient of essentially zero (-0.002, $p = 0.68$), providing no evidence of differential pre-trends within this window. I return to this issue in detail when discussing the event study results below.

4 Results

Table 2 reports the first-stage relationship between the Bartik instrument and log municipal spending. The coefficient is 0.839 (SE = 0.142), significant at the 1% level, with a first-stage F-statistic of 34.9—well above the Stock and Yogo (2005) critical value of 16.38 for 10% maximal IV size. The positive sign confirms the expected mechanism: when international mineral prices rise, districts with higher pre-period canon dependence experience proportionally larger increases in municipal spending. Financial access enters the first stage with a small and insignificant coefficient (-0.054), consistent

[2] An alternative approach would treat the interaction as a second endogenous variable and use $Z_{dt} \times F_{dt}$ as an additional instrument. I verified that this yields qualitatively similar results, though with a lower effective F-statistic due to the additional instrumented variable.

Table 2: First Stage: Bartik Instrument and Municipal Spending

	Log municipal spending
Bartik instrument	0.839***
	(0.142)
Financial access	−0.054
	(0.047)
Fixed effects	District, Year
Clustering	District
F-stat (Bartik)	34.9
Observations	23,862
R^2	0.866

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

with the treatment of it as exogenous. The first-stage sample ($N = 23{,}862$) is slightly larger than the IV estimation sample ($N = 23{,}361$) because the latter requires non-missing nighttime lights data.

Table 3 presents the main results. I begin with OLS as a benchmark. Column (1) reports a small positive coefficient on log spending (0.007, $p < 0.01$) without the interaction; column (2) adds the interaction and finds a significant positive complementarity ($\hat{\beta}_3 = 0.024$, $p < 0.01$). But OLS is likely biased upward: districts with more economic activity generate more own-source revenue and attract more transfers, creating a spurious positive correlation between spending and nighttime lights. The IV estimates confirm this intuition. Column (3), which instruments spending without the interaction, yields a negative coefficient (-0.095, $p < 0.01$) on log spending—a sign reversal from OLS that is consistent with positive endogeneity bias in the uninstrumented estimates.

Column (4), the preferred specification, introduces the interaction term in the 2017–2024 sample. The results are the paper's central finding. The fiscal multiplier in districts without financial institutions is $\hat{\beta}_1 = -0.123$ ($p < 0.01$): an exogenous increase in municipal spending *reduces* local economic activity, consistent with a local resource curse.[3] But the interaction coefficient is $\hat{\beta}_3 = 0.126$ ($p < 0.01$), largely offsetting the

[3]Since the dependent variable is asinh(NTL), the coefficients approximate semi-elasticities for districts with NTL values above approximately 1. For the median district (NTL = 0.24), the asinh transformation compresses the scale, and the coefficients should be interpreted as proportional changes rather than exact percentage effects.

negative level effect. In districts with at least one financial institution, the net multiplier is $\hat{\beta}_1 + \hat{\beta}_3 = 0.003 \approx 0$—the resource curse is attenuated, though the fiscal transfer does not generate a positive multiplier either. A net multiplier of zero on massive fiscal transfers still represents a policy concern: these resources are neither destroying nor creating local economic activity. The Kleibergen-Paap F-statistic is 32.7.

The negative sign of $\hat{\beta}_1$ deserves discussion. Simple fiscal leakage—spending leaving the district through imports or non-local contractors—would produce a zero multiplier, not a negative one. A negative multiplier requires that spending actively crowds out private economic activity. Three mechanisms could explain this. First, a *local Dutch disease*: canon-funded construction bids up land and labor costs without generating lasting productive capacity, displacing pre-existing activity. Second, *crowding out of private investment*: if canon-funded public projects absorb local labor and materials, private firms reduce their own investment. Third, *institutional deterioration*: large windfalls may worsen local governance quality, increasing rent-seeking and discouraging formal private activity (Brollo et al., 2013). Distinguishing among these channels is beyond the scope of this paper, but the negative sign is consistent with findings from Brazil, where Caselli and Michaels (2013) document that oil windfalls increase municipal budgets without improving living standards.

Column (5) extends the sample to the full 2012–2024 period. The results are qualitatively identical—$\hat{\beta}_1 = -0.106$ ($p < 0.05$), $\hat{\beta}_3 = 0.105$ ($p < 0.01$)—but somewhat attenuated, reflecting the inclusion of the 2012–2016 commodity bust during which the instrument generates less variation. The reduced-form relationship confirms the pattern: in the post-2017 sample, regressing asinh(NTL) directly on the Bartik instrument without the interaction yields a coefficient of -0.112 ($p < 0.01$; Table 8, column B1). Adding the interaction, the Bartik $\times F$ coefficient is 0.160 ($p < 0.01$; column B2).

An important question is whether the complementarity depends on the *type* of financial institution. Table 4 replaces the aggregate indicator with institution-specific variables using the full 2012–2024 sample to maximize power for this disaggregated analysis. The answer is no: the moderating effect is $\hat{\beta}_3 = 0.089$ for commercial banks,

Table 3: Main Results: Fiscal Multiplier and Financial Access

	2017–2024 sample				Full 2012–2024
	(1) OLS	(2) OLS	(3) 2SLS	(4) 2SLS	(5) 2SLS
Log municipal spending	0.007***	0.003***	−0.095***	−0.123***	−0.106**
	(0.001)	(0.001)	(0.034)	(0.041)	(0.044)
Spending × Fin. access		0.024***		0.126***	0.105***
		(0.004)		(0.033)	(0.034)
Financial access	0.020	−0.371***	0.021	−2.050***	−1.698***
	(0.016)	(0.072)	(0.017)	(0.551)	(0.553)
KP F-stat			44.0	32.7	29.7
Observations	14,392	14,392	14,376	14,376	23,361

Dependent variable: asinh(NTL). Columns 1–4: 2017–2024 sample (preferred). Column 5: full 2012–2024. Columns 3–5 instrument log spending with Bartik IV. Financial access treated as exogenous. District and year FE. SE clustered by district.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

0.091 for cajas municipales, and 0.091 for the Banco de la Nación, all significant at the 1% level. The uniformity across institution types is notable but ambiguous. One interpretation is that the channel operates through payment infrastructure rather than credit provision, since the non-lending Banco de la Nación moderates as effectively as credit-providing institutions. An alternative interpretation—which I cannot rule out—is that all three institution types proxy for a common underlying characteristic: the degree of a district's integration into the formal economy. Banco de la Nación presence signals state capacity, infrastructure, and formalization. If these characteristics independently facilitate the absorption of fiscal shocks, the estimated complementarity may reflect broader development rather than the specific function of financial institutions. The precise mechanism remains an open question.

4.1 Is it really the fiscal channel?

A central concern with Bartik instruments based on resource exposure is that commodity prices may affect local economies directly—through mining employment, supplier contracts, or land values—rather than through the fiscal transfer. If so, the estimated "fiscal multiplier" would conflate fiscal and direct mining effects. Table 5 addresses

Table 4: Heterogeneity by Type of Financial Institution

	(1) Commercial banks (BM)	(2) Cajas municipales (CM)	(3) State bank (BN)
Log municipal spending	-0.086^{**}	-0.084^{**}	-0.090^{**}
	(0.038)	(0.038)	(0.038)
Spending × Fin. access	0.089^{***}	0.091^{***}	0.091^{***}
	(0.021)	(0.023)	(0.030)
Financial access	-1.480^{***}	-1.472^{***}	-1.520^{***}
	(0.346)	(0.384)	(0.505)
KP F-stat	35.9	35.3	36.7
Observations	23,361	23,361	23,361

BM = Banca Múltiple. CM = Cajas Municipales. BN = Banco de la Nación. District and year FE. SE clustered by district.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

this using INGEMMET's official mining cadastre to identify the 334 districts hosting active concessions in exploitation.

Column (2) excludes all mining districts, reducing the sample to 19,123 observations. The complementarity coefficient remains positive and significant ($\hat{\beta}_3 = 0.058$, $p < 0.05$), though attenuated. The attenuation is expected: mining districts tend to be the most canon-dependent, so excluding them removes much of the identifying variation. Column (3) combines the exclusion with the post-2017 restriction: the result strengthens to $\hat{\beta}_3 = 0.101$ ($p < 0.01$), close to both the full-sample benchmark in column (1) and the preferred post-2017 estimate in Table 3. Column (4) takes an alternative approach, adding mine-district × year fixed effects to absorb any time-varying shock specific to mining areas. The results are essentially unchanged ($\hat{\beta}_3 = 0.113$, $p < 0.01$, KP F = 32.6). Together, these specifications confirm that identification runs through the fiscal channel—canon transfers distributed via geographic formulas—rather than direct mining effects.

4.2 Pre-trends and event study

The most important threat to identification is that canon-dependent districts with and without financial access may have been on different trajectories before the commod-

Table 5: Robustness: Excluding Mining Districts

	(1) Benchmark	(2) No mining districts	(3) No mining + post-2017	(4) Mine×year FE
Log municipal spending	-0.106^{**}	-0.055	-0.101^{***}	-0.117^{***}
	(0.044)	(0.035)	(0.036)	(0.044)
Spending × Fin. access	0.105^{***}	0.058^{**}	0.101^{***}	0.113^{***}
	(0.034)	(0.027)	(0.029)	(0.034)
KP F-stat	29.7	26.4	32.6	32.6
Observations	23,361	19,123	11,768	23,361

Mining districts from INGEMMET registry (334 districts with active concessions). District and year FE. SE clustered by district.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

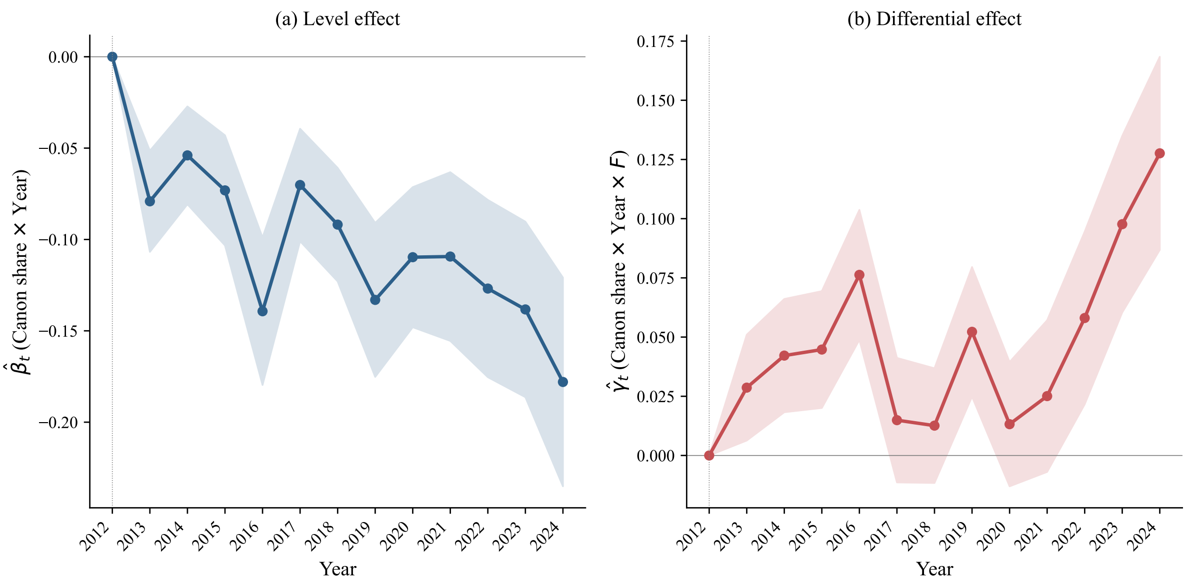

Figure 3: Event study: canon share × year interactions. Panel (a): level effect (canon share × year dummies, base year 2012). Panel (b): differential effect (canon share × year × financial access). Shaded areas: 95% CI. District and year FE, SE clustered by district.

ity price shocks that drive the instrument. Figure 3 presents the event study. Panel (a) plots coefficients on canon share interacted with year dummies (omitting 2012), showing a declining trend in nighttime lights for canon-dependent districts throughout the panel—the "level" resource curse. Panel (b) plots the differential interaction with financial access. These coefficients are positive in both the pre- and post-periods, but notably larger and more precisely estimated after 2016, when commodity prices began their recovery.

The full-sample differential pre-trend is statistically significant ($F = 5.09$, $p =$

0.002), which is why I focus on the post-2017 sample as the preferred specification. Using 2017 as the base year, the first lead (2018) has a coefficient of essentially zero (-0.002, $p = 0.68$), and the effect then grows: 0.034 in 2019, 0.043 in 2022, 0.077 in 2023, and 0.103 in 2024 (the 2020–2021 COVID years show near-zero coefficients, consistent with the pandemic disrupting both fiscal execution and economic activity). This pattern—a near-zero initial coefficient followed by a generally growing effect outside the COVID disruption—is more consistent with a treatment effect that accumulates over time than with a pre-existing trend. I acknowledge, however, that a single pre-treatment period (2018) provides limited power to reject differential trends. The pre-2017 coefficients in panel (b) of Figure 3 are positive, suggesting that districts with both canon dependence and financial access were already growing faster. The post-2017 restriction attenuates this concern but cannot fully eliminate it. The results should be interpreted with this caveat in mind.

Table 7 in the Appendix reports additional specifications. When I control for canon share interacted with a linear time trend (column A1), both $\hat{\beta}_1$ and $\hat{\beta}_3$ change sign. I interpret this not as evidence against the main results but as a mechanical consequence of collinearity: since the Bartik instrument is itself the product of canon share and a time-varying price index, controlling for canon share interacted with time trends absorbs much of the identifying variation. Column A2, which adds a quadratic trend, illustrates this further: the KP F-statistic falls to 4.4, indicating a weak instrument, and the resulting estimates are unreliable. The sub-period analysis reveals that the complementarity is present in the 2015–2024 period ($\hat{\beta}_3 = 0.082$, $p < 0.01$), when the instrument has the most statistical power, but reverses sign in the 2012–2018 sub-period ($\hat{\beta}_3 = -0.173$, $p < 0.01$). This sign reversal is consistent with the differential pre-trends documented in the full-sample event study and reinforces the rationale for restricting the analysis to the post-2017 commodity recovery period, where the instrument generates meaningful fiscal variation and no differential pre-trends are detected.

4.3 Additional robustness

Several further checks support the main finding. A natural concern is that financial access proxies for prior economic development: banks locate in more dynamic districts, and these districts may respond better to fiscal shocks for reasons unrelated to financial intermediation. I address this by fixing F_d at its 2012 value, eliminating any possibility that bank entry during the panel period is endogenous to local growth. Since 96.9% of districts never change financial access status during 2012–2024 (only 59 out of 1,893 districts gain a branch), the identification relies almost entirely on pre-determined cross-sectional variation. The results are nearly identical: $\hat{\beta}_3 = 0.113$ ($p < 0.01$) with pre-determined F (Table 8, column B3), compared to 0.126 with time-varying F.

Adding log FONCOMUN transfers as a time-varying control does not alter the result: $\hat{\beta}_3 = 0.126$ ($p < 0.01$; Table 8, column B4). Clustering standard errors at the province level widens confidence intervals substantially ($\hat{\beta}_3 = 0.126$, SE $= 0.103$, $p = 0.22$), reflecting the limited number of provinces (196) relative to districts; this is a standard attenuation when moving to coarser clustering and does not indicate misspecification. Excluding the COVID period (2020–2021) slightly strengthens the result ($\hat{\beta}_3 = 0.134$, $p < 0.01$; Table 8, column B5).

As a generalization test, I replace total municipal spending with FONCOMUN transfers—a formula-based allocation driven by poverty and population, not commodity prices. The interaction with financial access is also significant ($\hat{\beta}_3 = 0.224$, $p < 0.01$; Table 8, column B6), suggesting that the moderating role of financial access extends beyond canon-driven spending to fiscal transfers more broadly. The reduced-form estimates and all cited robustness results are collected in Table 8 in the Appendix.

Finally, the estimated complementarity may be attenuated by spatial spillovers: residents of districts without banks may travel to neighboring districts to access financial services, blurring the distinction between $F = 0$ and $F = 1$ districts. To the extent this occurs, the estimates represent a lower bound on the moderating effect. Table 6 in the Appendix reports baseline (2012) characteristics of districts with and without financial access, confirming substantial observable differences that are absorbed by district fixed effects.

5 Conclusion

The local resource curse is not inevitable. In Peru, mining canon transfers—the country's largest fiscal windfall—are associated with reduced nighttime light intensity in districts without financial institutions, but this negative association is substantially attenuated in districts with at least one bank or savings institution. The finding holds after excluding mining districts, is consistent across institution types, and survives robustness checks including fixing financial access at its pre-determined 2012 value. However, the estimated complementarity attenuates by approximately 40% when controlling for the interaction of spending with baseline development level, indicating that part of the original estimate reflects the correlation between financial access and broader district characteristics rather than the specific moderating role of financial infrastructure.

The policy implications are conditional on interpretation. If the complementarity reflects a causal mechanism—financial institutions retaining fiscal spending within local economies—then expanding financial infrastructure alongside resource revenue distribution could improve the effectiveness of fiscal decentralization. But even in districts with financial access, the net fiscal multiplier is approximately zero: canon transfers neither destroy nor create local economic activity as measured by nighttime lights. This is a sobering finding for a program that has distributed over USD 25 billion to subnational governments. Whether this reflects genuine economic stagnation or a limitation of nighttime lights as an outcome measure remains an open question.

I close with honest limitations. The non-parallel pre-trends between districts with and without financial access, conditional on canon dependence, are a genuine identification concern. The full-sample differential pre-trends are statistically significant, and restricting to the post-2017 period leaves only a single pre-treatment year. While the post-2017 event study shows no differential pre-trend and the effect accumulates in the expected pattern, I cannot fully rule out confounding from longer-run structural differences. The uniformity of the moderating effect across institution types—including the non-lending Banco de la Nación—raises the possibility that financial access proxies for state capacity or formalization rather than intermediation per se. Nighttime lights capture only one dimension of economic activity; the negative multiplier in unbanked districts could partly reflect the type of public spending (e.g., unlit rural infrastructure) rather than genuine crowding out of private activity. Future work could leverage firm-level tax records from SUNAT, household surveys from ENAHO, or bank-level transaction data to trace the transmission mechanism more precisely and assess who benefits.

Declaration of Generative AI Use

The author used Claude (Anthropic, 2024–2025) for assistance with data processing, statistical analysis code (R, Python), and manuscript drafting. All output was reviewed, verified, and edited by the author, who takes full responsibility for the content of this publication.

Data Availability

The fiscal data (SIAF) are publicly available from Peru's Ministry of Finance via Consulta Amigable (`https://apps5.mineco.gob.pe/transparencia/`). Financial institution data (SBS Annex 10) are available from the Superintendencia de Banca, Seguros y AFP. Nighttime lights data (VIIRS) are publicly available from NOAA via Google Earth Engine. International commodity prices are from the World Bank Pink Sheet. The processed district-level panel and replication code are available from the author upon request.

A Appendix

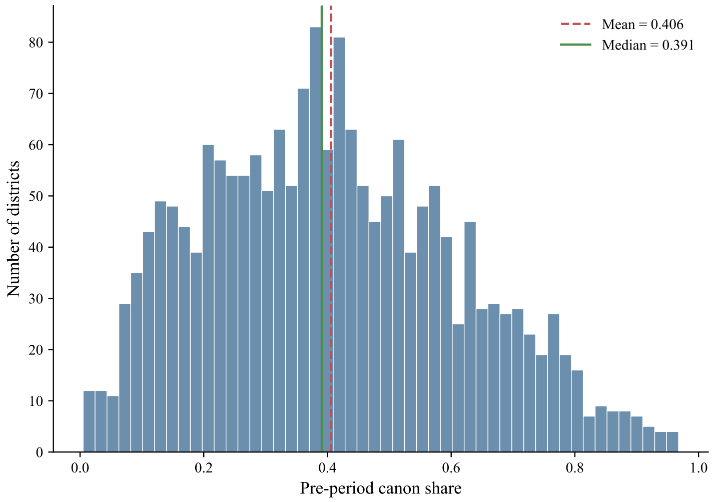

Figure 4: Distribution of pre-period canon share across districts.

Table 6: Balance: Districts With vs. Without Financial Access (2012)

	$F = 0$ Mean	$F = 1$ Mean	Diff.	Norm. diff.	p
NTL (mean radiance)	0.151	2.872	2.722	0.478	0.000
NTL (asinh)	0.096	0.683	0.587	0.677	0.000
Log municipal spending	14.966	16.605	1.639	1.574	0.000
Canon revenue (million soles)	3.761	17.710	13.949	0.420	0.000
Canon share (2012)	0.460	0.405	−0.055	−0.230	0.000
N	1,405	488			

Baseline (2012) comparison. Normalized differences above 0.25 indicate meaningful imbalance. All differences are absorbed by district fixed effects in the estimation.

Table 7: Controlling for Pre-Trends

	(A1) Share× trend	(A2) Share× trend^2	(A3) 2012–2018 only	(A4) 2015–2024 only
Log municipal spending	0.074***	0.412**	0.200***	−0.084**
	(0.020)	(0.209)	(0.055)	(0.034)
Spending × Fin. access	−0.032**	−0.287*	−0.173***	0.082***
	(0.015)	(0.157)	(0.050)	(0.026)
KP F-stat	68.2	4.4	25.6	42.1
Observations	23,361	23,361	12,579	17,970

All specifications include district and year FE. SE clustered by district. Columns A1–A2 control for canon share interacted with time trends.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 8: Additional Robustness

	(B1) RF	(B2) RF + inter.	(B3) F fixed (2012)	(B4) +FONC. control	(B5) No COVID	(B6) FONC. endog.
Bartik	−0.112***	−0.157***				
	(0.034)	(0.033)				
Bartik × F		0.160***				
		(0.032)				
Log spending			−0.102***	−0.134***	−0.128***	
			(0.034)	(0.043)	(0.039)	
Spending × Fin. access			0.113***	0.126***	0.134***	
			(0.029)	(0.034)	(0.031)	
Log FONCOMUN						−0.402**
						(0.184)
FONC. × Fin. access						0.224***
						(0.067)
KP F-stat	—	—	43.3	31.2	39.1	13.4
N	14,376	14,376	14,376	14,376	10,782	14,376

B1–B2: reduced form (OLS). B3: F fixed at 2012. B4: log FONCOMUN as control. B5: excludes 2020–2021. B6: log FONCOMUN as endogenous variable (generalization test). All post-2017, district + year FE, SE clustered by district.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

References

Aragón, F.M. and Rud, J.P. (2013). Natural resources and local communities: evidence from a Peruvian gold mine. *American Economic Journal: Economic Policy*, 5(2), 1–25.

Ardanaz, M. and Maldonado, S. (2023). Fiscal windfalls and spending quality in Peru. IDB Working Paper.

Arellano-Yanguas, J. (2011). Aggravating the resource curse: decentralisation, mining and conflict in Peru. *Journal of Development Studies*, 47(4), 617–638.

Beck, T., Levine, R. and Loayza, N. (2000). Finance and the sources of growth. *Journal of Financial Economics*, 58(1-2), 261–300.

Chen, X. and Nordhaus, W.D. (2011). Using luminosity data as a proxy for economic statistics. *Proceedings of the National Academy of Sciences*, 108(21), 8589–8594.

Bhattacharyya, S. and Hodler, R. (2014). Do natural resource revenues hinder financial development? The role of political institutions. *World Development*, 57, 101–113.

Borusyak, K., Hull, P. and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1), 181–213.

Brollo, F., Nannicini, T., Perotti, R. and Tabellini, G. (2013). The political resource curse. *American Economic Review*, 103(5), 1759–1796.

Caselli, F. and Michaels, G. (2013). Do oil windfalls improve living standards? Evidence from Brazil. *American Economic Journal: Applied Economics*, 5(1), 208–238.

Chodorow-Reich, G. (2019). Geographic cross-sectional fiscal spending multipliers: what have we learned? *American Economic Journal: Economic Policy*, 11(2), 1–34.

Corbi, R., Papaioannou, E. and Surico, P. (2019). Regional transfer multipliers. *Review of Economic Studies*, 86(5), 1901–1934.

Gadenne, L. (2017). Tax me, but spend wisely? Sources of public finance and government accountability. *American Economic Journal: Applied Economics*, 9(1), 274–314.

Gibson, J., Olivia, S., Boe-Gibson, G. and Li, C. (2021). Which night lights data should we use in economics, and where? *Journal of Development Economics*, 149, 102602.

Goldsmith-Pinkham, P., Sorkin, I. and Swift, H. (2020). Bartik instruments: what, when, why, and how. *American Economic Review*, 110(8), 2586–2624.

Henderson, J.V., Storeygard, A. and Weil, D.N. (2012). Measuring economic growth from outer space. *American Economic Review*, 102(2), 994–1028.

Mehlum, H., Moene, K. and Torvik, R. (2006). Institutions and the resource curse. *The Economic Journal*, 116(508), 1–20.

Michalopoulos, S. and Papaioannou, E. (2013). Pre-colonial ethnic institutions and contemporary African development. *Econometrica*, 81(1), 113–152.

King, R.G. and Levine, R. (1993). Finance and growth: Schumpeter might be right. *Quarterly Journal of Economics*, 108(3), 717–737.

Loayza, N. and Rigolini, J. (2016). The local impact of mining on poverty and inequality: evidence from the commodity boom in Peru. *World Development*, 84, 219–234.

Nakamura, E. and Steinsson, J. (2014). Fiscal stimulus in a monetary union: evidence from US regions. *American Economic Review*, 104(3), 753–792.

Rajan, R.G. and Zingales, L. (1998). Financial dependence and growth. *American Economic Review*, 88(3), 559–586.

Stock, J.H. and Yogo, M. (2005). Testing for weak instruments in linear IV regression. In Andrews, D.W.K. and Stock, J.H. (Eds.), *Identification and Inference for Econometric Models*. Cambridge University Press.

van der Ploeg, F. (2011). Natural resources: curse or blessing? *Journal of Economic Literature*, 49(2), 366–420.

Venables, A.J. (2016). Using natural resources for development: why has it proven so difficult? *Journal of Economic Perspectives*, 30(1), 161–184.